Assignment 3

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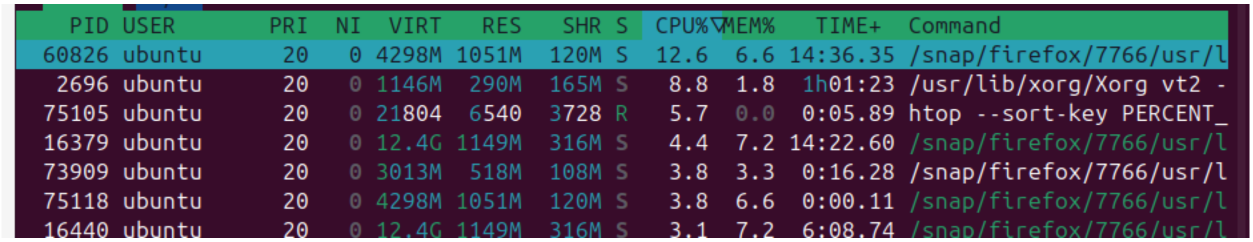
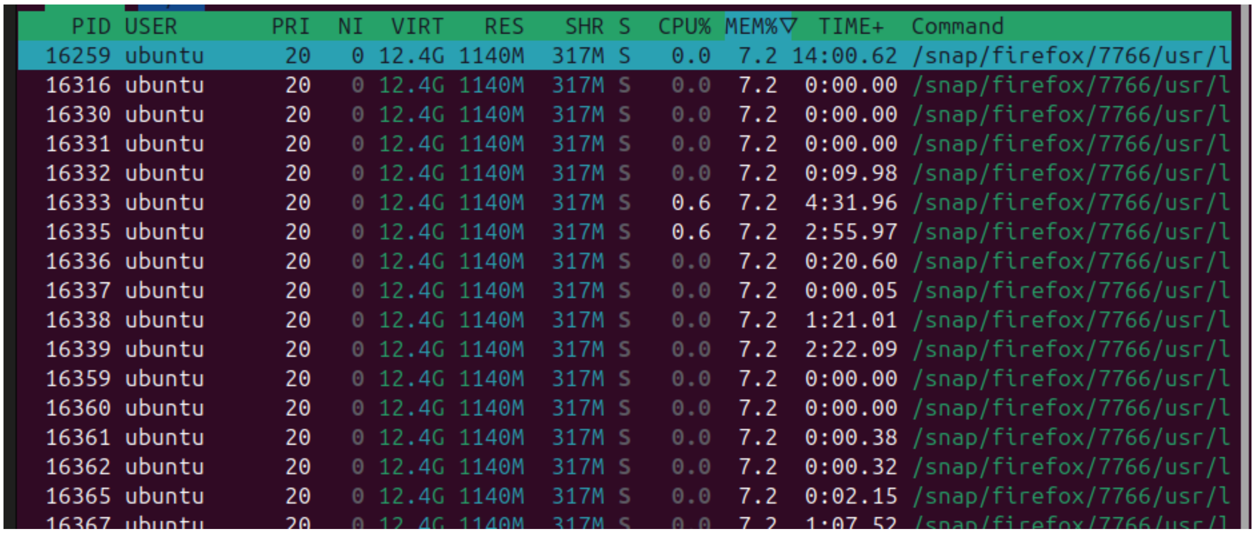
Repo link: https://github.com/kaiyi03/CWM-Project/tree/main/assignment3

Q1



PID	USER	PRI	NI	VIRT	RES	SHR	S	CPU%▽	MEM%	TIME+	Command
60826	ubuntu	20	0	4298M	1051M	120M	S	12.6	6.6	14:36.35	/snap/firefox/7766/usr/l
2696	ubuntu	20	0	1146M	290M	165M	S	8.8	1.8	1h01:23	/usr/lib/xorg/Xorg vt2 -
75105	ubuntu	20	0	21804	6540	3728	R	5.7	0.0	0:05.89	htop --sort-key PERCENT_
16379	ubuntu	20	0	12.4G	1149M	316M	S	4.4	7.2	14:22.60	/snap/firefox/7766/usr/l
73909	ubuntu	20	0	3013M	518M	108M	S	3.8	3.3	0:16.28	/snap/firefox/7766/usr/l
75118	ubuntu	20	0	4298M	1051M	120M	S	3.8	6.6	0:00.11	/snap/firefox/7766/usr/l
16440	ubuntu	20	0	12.4G	1149M	316M	S	3.1	7.2	6:08.74	/snap/firefox/7766/usr/l

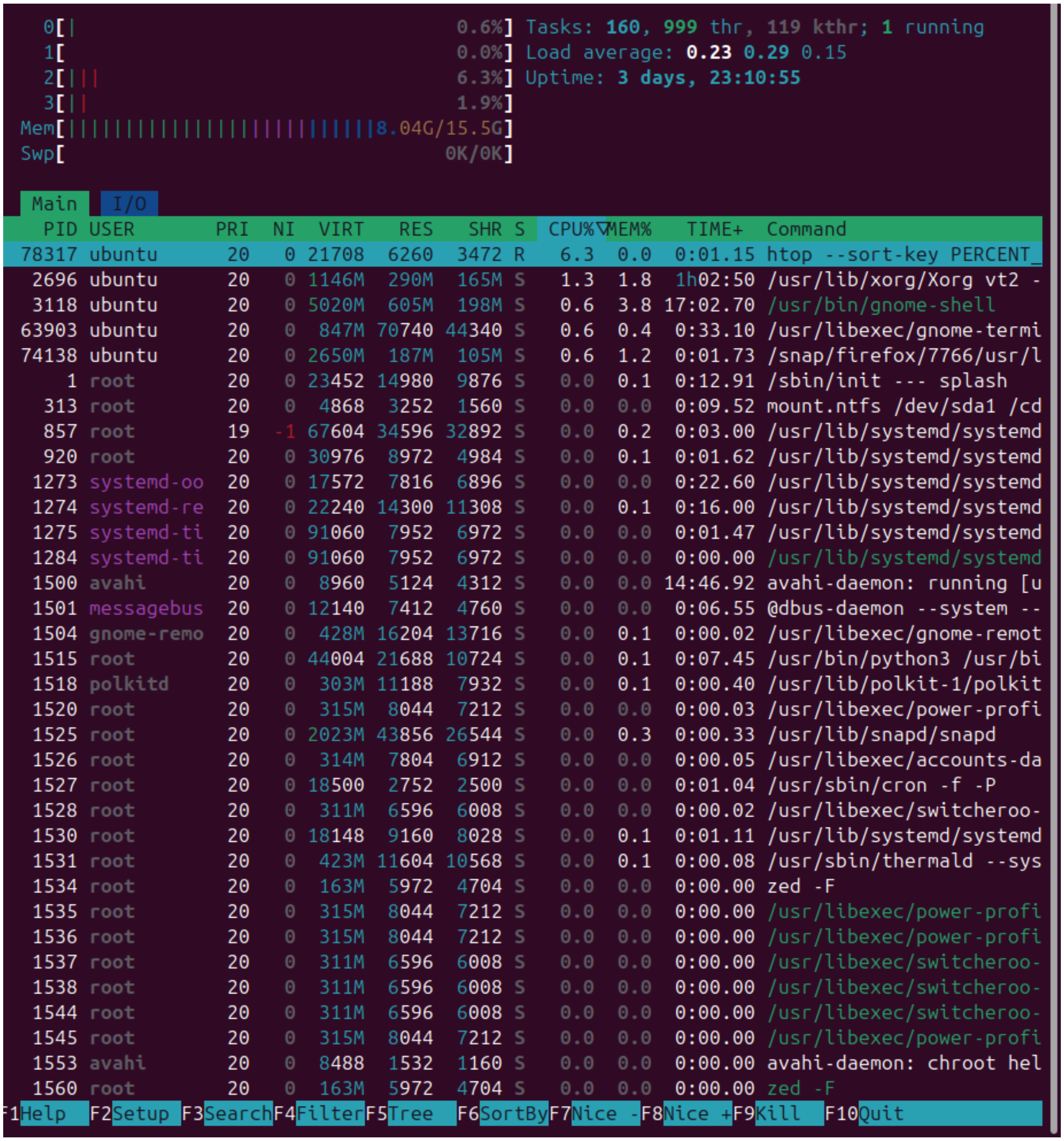


PID	USER	PRI	NI	VIRT	RES	SHR	S	CPU%	MEM%▽	TIME+	Command
16259	ubuntu	20	0	12.4G	1140M	317M	S	0.0	7.2	14:00.62	/snap/firefox/7766/usr/l
16316	ubuntu	20	0	12.4G	1140M	317M	S	0.0	7.2	0:00.00	/snap/firefox/7766/usr/l
16330	ubuntu	20	0	12.4G	1140M	317M	S	0.0	7.2	0:00.00	/snap/firefox/7766/usr/l
16331	ubuntu	20	0	12.4G	1140M	317M	S	0.0	7.2	0:00.00	/snap/firefox/7766/usr/l
16332	ubuntu	20	0	12.4G	1140M	317M	S	0.0	7.2	0:09.98	/snap/firefox/7766/usr/l
16333	ubuntu	20	0	12.4G	1140M	317M	S	0.6	7.2	4:31.96	/snap/firefox/7766/usr/l
16335	ubuntu	20	0	12.4G	1140M	317M	S	0.6	7.2	2:55.97	/snap/firefox/7766/usr/l
16336	ubuntu	20	0	12.4G	1140M	317M	S	0.0	7.2	0:20.60	/snap/firefox/7766/usr/l
16337	ubuntu	20	0	12.4G	1140M	317M	S	0.0	7.2	0:00.05	/snap/firefox/7766/usr/l
16338	ubuntu	20	0	12.4G	1140M	317M	S	0.0	7.2	1:21.01	/snap/firefox/7766/usr/l
16339	ubuntu	20	0	12.4G	1140M	317M	S	0.0	7.2	2:22.09	/snap/firefox/7766/usr/l
16359	ubuntu	20	0	12.4G	1140M	317M	S	0.0	7.2	0:00.00	/snap/firefox/7766/usr/l
16360	ubuntu	20	0	12.4G	1140M	317M	S	0.0	7.2	0:00.00	/snap/firefox/7766/usr/l
16361	ubuntu	20	0	12.4G	1140M	317M	S	0.0	7.2	0:00.38	/snap/firefox/7766/usr/l
16362	ubuntu	20	0	12.4G	1140M	317M	S	0.0	7.2	0:00.32	/snap/firefox/7766/usr/l
16365	ubuntu	20	0	12.4G	1140M	317M	S	0.0	7.2	0:02.15	/snap/firefox/7766/usr/l
16367	ubuntu	20	0	12.4G	1140M	317M	S	0.0	7.2	1:07.52	/snap/firefox/7766/usr/l

Firefox seems to be at the top CPU and MEM consumption

Running a compute heavy program will put you at the top. (e.g., opening multiple tabs, running it all simultaneously)

Q2.



Looking at the 4 cores, we can approximate the idle time for each core

Core 0: approx 99.4%

Core 1: 100%

Core 2: 93.7%

Core 3: 98.1%

Q3.

```
ubuntu@ubuntu:~/CWM-Project/assignment3$ sudo turbostat -q -S --show PkgWatt -i 1
PkgWatt
2.44
2.66
2.59
2.98
2.46
5.78
7.11
2.66
7.09
4.67
9.70
2.56
4.66
4.04
2.45
2.55
2.62
2.57
2.66
5.51
2.99
3.13
2.67
```

```
ubuntu@ubuntu:~/CWM-Project/assignment3$ sudo turbostat -q -S --Joules --show Pkg_J -i 1
Pkg_J
2.65
2.54
2.56
2.62
10.22
9.71
8.03
8.13
5.69
6.65
4.75
5.78
7.08
3.39
4.81
7.05
8.62
8.57
5.46
6.05
6.47
4.04
3.61
4.38
3.66
```

Yes, the baseline fluctuates slightly. Even at idle, the CPU handles background tasks and timer events, causing small spikes in power used. However, the variation is small, and the average seems to remain relatively stable.

Q4.

```
ubuntu@ubuntu:~/CWM-Project/assignment3$ sudo turbostat -q -S --show Busy%,Bzy_M
Hz,PkgWatt,CorWatt,PkgTmp,CoreTmp -i 1
Busy%   Bzy_MHz CoreTmp PkgTmp  PkgWatt CorWatt
25.86   3600    48      48      8.66    6.66
7.31    3598    38      38      4.24    2.41
1.72    3592    38      38      2.80    1.02
27.77   3599    40      40      9.44    7.48
20.25   3598    39      39      7.26    5.33
3.48    3598    38      38      3.33    1.53
4.63    3592    38      38      3.62    1.82
7.47    3596    37      37      4.29    2.46
```

turbostat's Busy% gives a more precise measurement of average CPU activity than ps/htop.

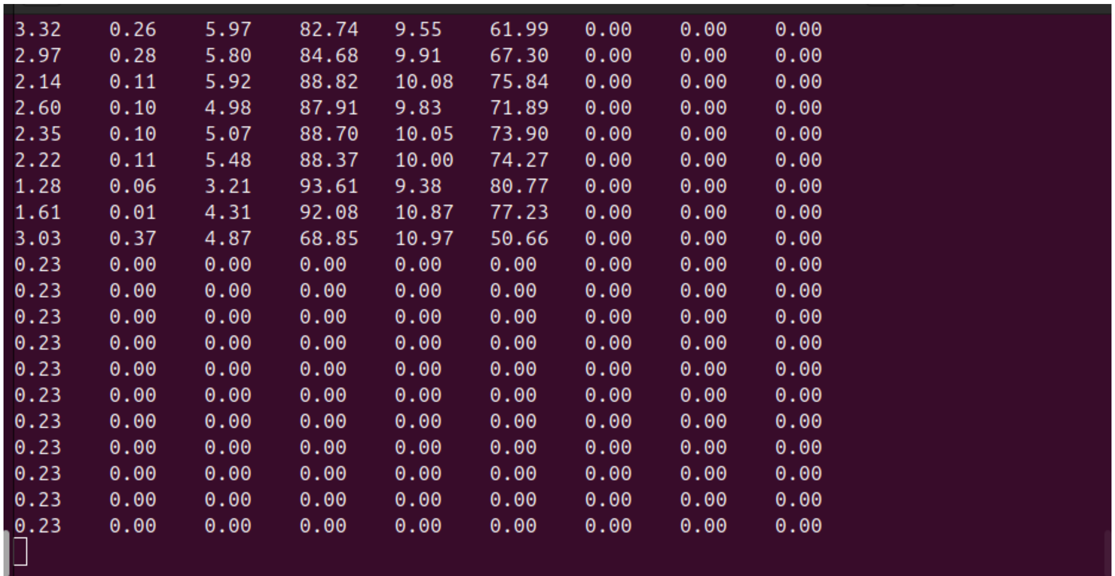
ps/htop shows average CPU usage for processes that are running, turbostat may show a slightly higher or more variable Busy% because it captures all hardware activity tasks that ps do not capture.

turbostat gives better insight into true CPU utilisation

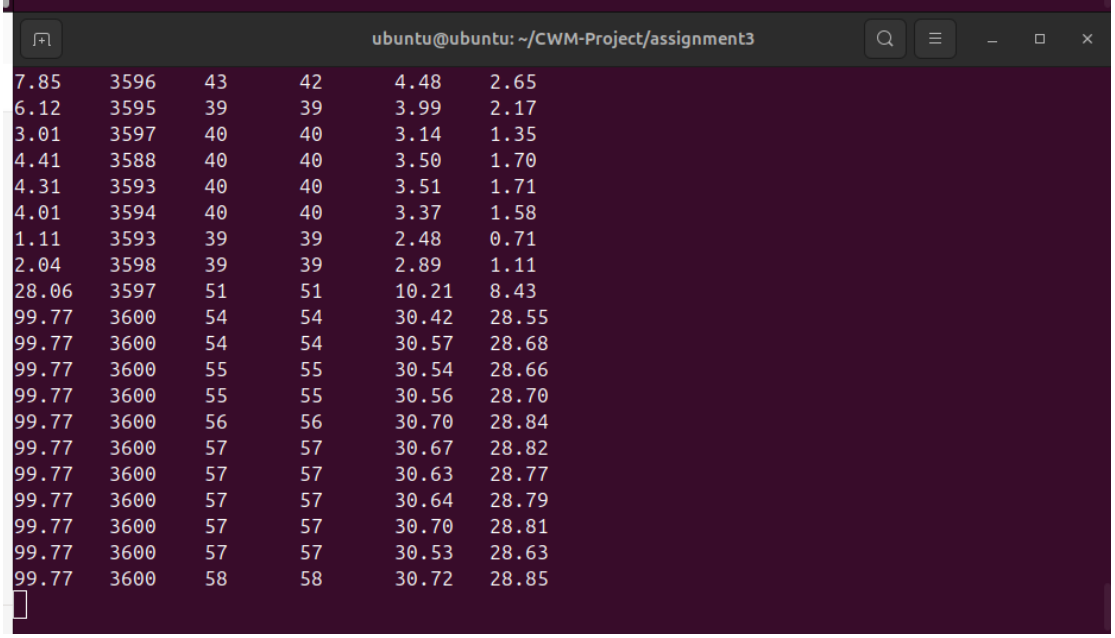
Q5.

```
ubuntu@ubuntu:~/CWM-Project/assignment3$ sudo turbostat -q -S --show Pkg%pc2,Pkg
%pc3,Pkg%pc6,Pkg%pc7,Pkg%pc8,CPU%c1,CPU%c3,CPU%c6,CPU%c7 -i 1
CPU%c1  CPU%c3  CPU%c6  CPU%c7  Pkg%pc2 Pkg%pc3 Pkg%pc6 Pkg%pc7 Pkg%pc8
1.60    0.05    3.05    94.20   11.39   79.71   0.00    0.00    0.00
1.28    0.03    2.36    90.91   9.04    73.38   0.00    0.00    0.00
1.39    0.04    2.88    94.32   10.81   79.71   0.00    0.00    0.00
1.82    0.05    5.45    90.44   10.37   77.06   0.00    0.00    0.00
1.31    0.02    3.33    94.29   10.37   81.46   0.00    0.00    0.00
1.28    0.03    3.42    94.26   9.77    82.64   0.00    0.00    0.00
6.01    0.51    5.72    66.62   5.22    36.64   0.00    0.00    0.00
4.57    0.25    6.43    78.11   8.95    60.32   0.00    0.00    0.00
7.35    0.88    7.19    60.96   6.32    40.47   0.00    0.00    0.00
1.64    0.08    3.36    92.74   10.47   79.37   0.00    0.00    0.00
1.33    0.04    2.54    95.24   11.15   81.46   0.00    0.00    0.00
4.28    0.25    4.93    76.20   6.20    49.08   0.00    0.00    0.00
5.54    0.51    9.18    69.55   8.05    40.34   0.00    0.00    0.00
4.38    0.34    6.45    72.29   7.34    47.87   0.00    0.00    0.00
4.98    0.56    8.00    60.62   6.70    37.85   0.00    0.00    0.00
5.88    0.45    9.36    65.23   7.54    40.56   0.00    0.00    0.00
5.38    0.67    9.51    67.06   7.24    44.65   0.00    0.00    0.00
```

Q6a.



3.32	0.26	5.97	82.74	9.55	61.99	0.00	0.00	0.00
2.97	0.28	5.80	84.68	9.91	67.30	0.00	0.00	0.00
2.14	0.11	5.92	88.82	10.08	75.84	0.00	0.00	0.00
2.60	0.10	4.98	87.91	9.83	71.89	0.00	0.00	0.00
2.35	0.10	5.07	88.70	10.05	73.90	0.00	0.00	0.00
2.22	0.11	5.48	88.37	10.00	74.27	0.00	0.00	0.00
1.28	0.06	3.21	93.61	9.38	80.77	0.00	0.00	0.00
1.61	0.01	4.31	92.08	10.87	77.23	0.00	0.00	0.00
3.03	0.37	4.87	68.85	10.97	50.66	0.00	0.00	0.00
0.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00



ubuntu@ubuntu: ~/CWM-Project/assignment3

7.85	3596	43	42	4.48	2.65
6.12	3595	39	39	3.99	2.17
3.01	3597	40	40	3.14	1.35
4.41	3588	40	40	3.50	1.70
4.31	3593	40	40	3.51	1.71
4.01	3594	40	40	3.37	1.58
1.11	3593	39	39	2.48	0.71
2.04	3598	39	39	2.89	1.11
28.06	3597	51	51	10.21	8.43
99.77	3600	54	54	30.42	28.55
99.77	3600	54	54	30.57	28.68
99.77	3600	55	55	30.54	28.66
99.77	3600	55	55	30.56	28.70
99.77	3600	56	56	30.70	28.84
99.77	3600	57	57	30.67	28.82
99.77	3600	57	57	30.63	28.77
99.77	3600	57	57	30.64	28.79
99.77	3600	57	57	30.70	28.81
99.77	3600	57	57	30.53	28.63
99.77	3600	58	58	30.72	28.85

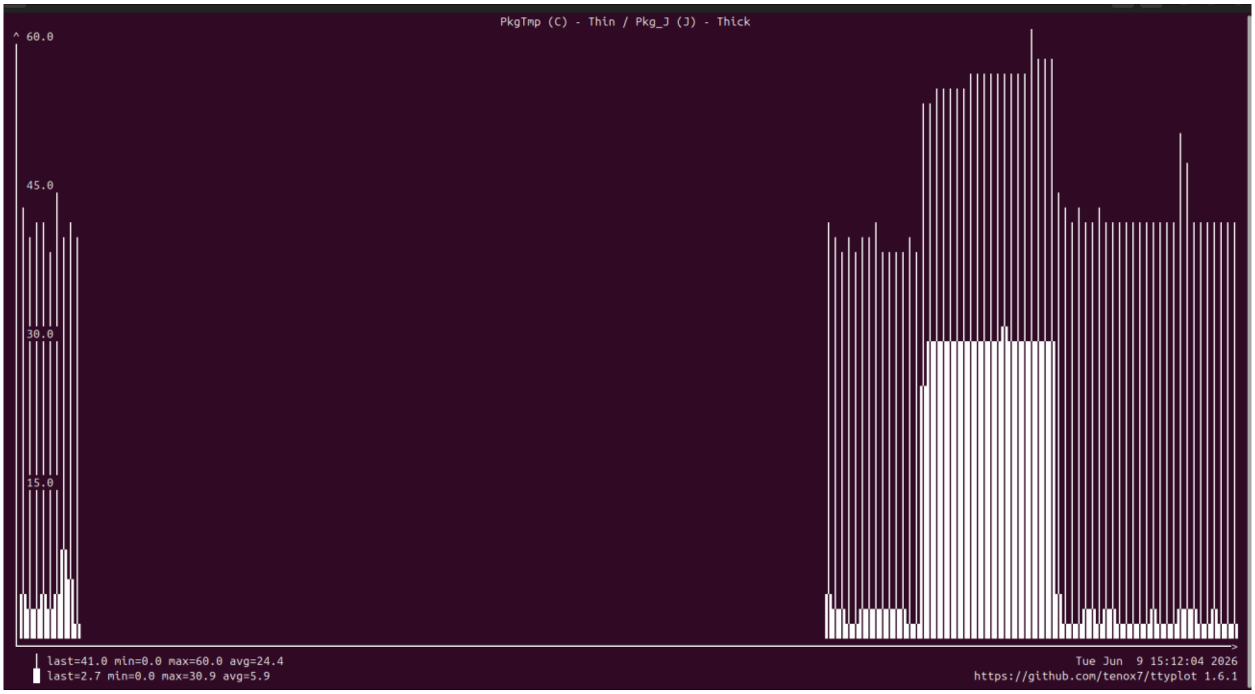
Busy% jumps significantly to 99.77

Pkgwatt increases significantly

Pkgcoretemp increases significantly too

C6 and c7 represents deep sleep and deeper sleep. When stress ng is run, these drop to 0 as the cores have no time to be idle.

Q6b.



When stress-ng is running we can see a spike in the pkg temp and pkgJ – both temp and power consumed increases.

Q6c.

At baseline the CPU drew approximately 5-6W (from q4)

During stress testing across 4 cores power peaked at approximately 30W.

If the machine sits idle for most of the day at baseline power, underutilisation means it consumes significant energy doing nothing.

Q7

Matmul_slow:

```
Pkg_J
207.30
207.30
```

Matmul_fast:

```
Pkg_J
548.70
548.70
```

Intuition tells you that Matmulslow should take longer than mat_mulfast so should consume more energy.

However, mat_mulfast which calls matmulfast3 consumes more energy perhaps because the function matmulfast3 needs to first transpose the matrix. This may add significant starting cost.

Q8.

At baseline yes, i observe cores spending significant time in deep sleep from Q5.

During stress-ng thjese drop to near 0 as cores were fully active

Drawback of deeper sleep states is longer wake up time needed.

Applications that might suffer could be things like financial systems that require low latency.

Q9.

To reduce idle power draw we may want to concentrate the work onto fewer fully utilised machines.

Wehn designing large data centres, heat becomes an important factor. Each cpu heating high temperature would require active cooling. This constraint would mean data centres would benefit from being built in places with colder climates.

Other modern accelerators like GPU offer better performance per watt hence consumig less energy.

Q10a.

1GB = 8 gigabits

Rate at 1Gbit/s

Hence time taken 8s

Q10b.

E = pt

E = 0.3 * 8 = 2.4J (low est)

Or 8.0J (high est)

Q10c.

If the data travels through multiple intermediate machines each step consumes additional energy. Assuming each intermediate device draws similar power, total network energy increases with more steps.

To make this more accurate we need to know the number of steps between source and destination, power draw of each intermediate device, whether intermediate links run at lower speeds as this which would increase transmission time and therefore energy.

Q11a. https://huggingface.co/datasets/RyokoAI/ShareGPT52K/tree/main

Q11b.

```
8.106098 sec
Pkg_J
88.95
88.95
```

Q11c.

Theoretically at full 1Gb/s the download should take 8 seconds consuming 2.4–8.0 J from the network interface alone. The measured download took 8.1 seconds which is very close to the theoretical time suggesting the network was operating near its full 1Gb/s rate.

However, the measured energy was 88.95 J, much higher than the theoretical 2.4–8.0 J.

Theoretical only measures the network port power. Turboset captures all other additional processes that happens during the download.

Q12.

480p

```
12.19
15.44
21.26
18.41
14.22
11.89
9.50
9.96
12.08
13.06
13.29
6.91
6.61
17.89
12.30
10.76
6.83
6.48
7.01
```

1080p

```
10.34
11.72
16.49
12.40
10.37
14.02
15.27
10.53
11.22
10.94
15.38
16.74
15.50
11.83
10.58
10.70
10.57
10.91
13.15
10.37
10.22
10.06
11.28
14.79
```

2160p

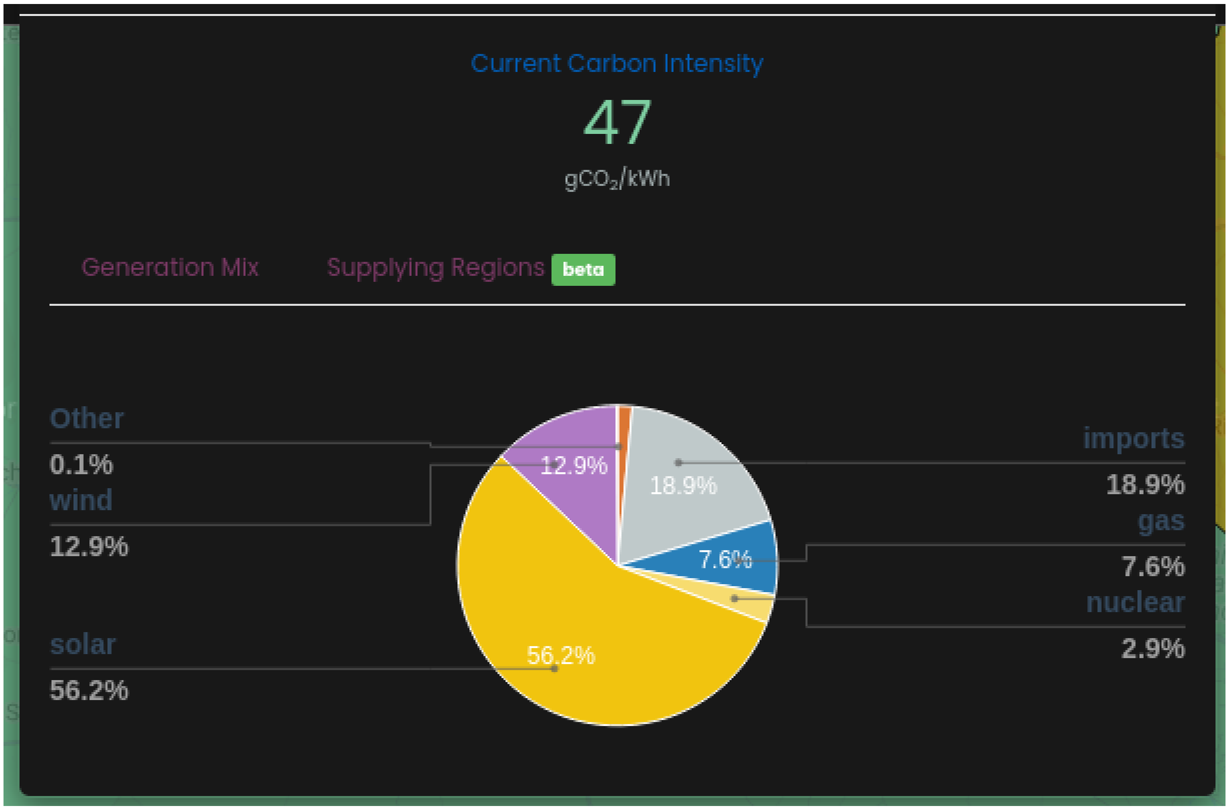
```
16.15
22.92
24.15
22.98
24.54
24.81
18.39
14.42
14.08
13.27
14.48
14.67
14.18
14.52
16.46
14.89
15.59
18.64
19.11
20.13
19.88
18.92
19.31
```

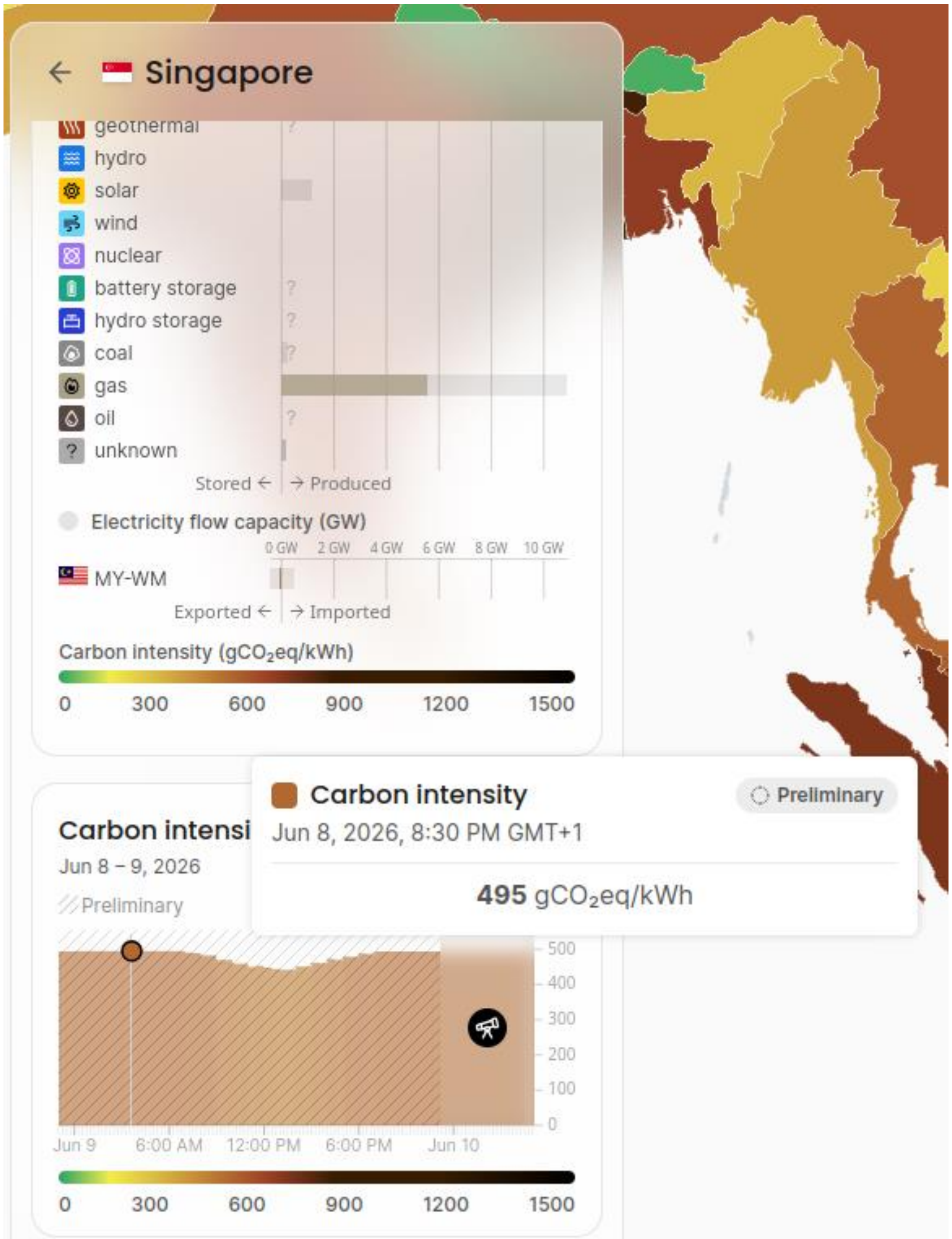
Higher resolution videos consume more energy.

This is because higher resolution video requires the CPU to decode more data per second, increasing energy consumption. This suggests that streaming at unnecessarily high resolutions has energy costs.

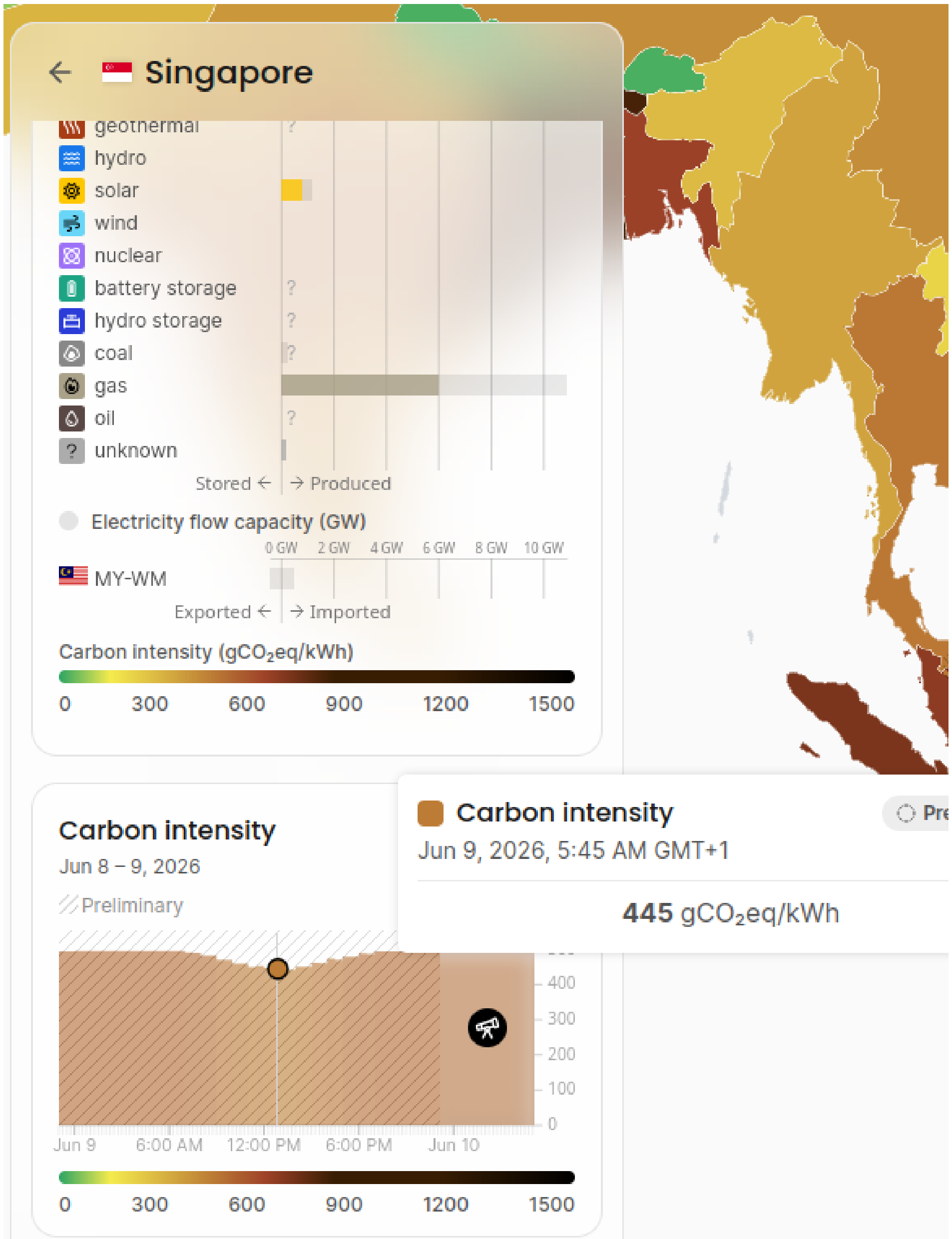
Q13.

Carbon intensity in Oxford





Singapore
geothermal
hydro
solar
wind
nuclear
battery storage
hydro storage
coal
gas
oil
unknown
Stored ← → Produced
Electricity flow capacity (GW)
0 GW 2 GW 4 GW 6 GW 8 GW 10 GW
MY-WM
Exported ← → Imported
Carbon intensity (gCO_2eq/kWh)
0 300 600 900 1200 1500
Carbon intensi
Jun 8 – 9, 2026
Preliminary
Carbon intensity
Preliminary
Jun 8, 2026, 8:30 PM GMT+1
495 gCO_2eq/kWh
500 400 300 200 100 0
Jun 9 6:00 AM 12:00 PM 6:00 PM Jun 10
0 300 600 900 1200 1500



Matmul_slow: 207.3J = 0.000057583kwh

Matmul_fast: 548.7J = 0.000152417kwh

CFP for oxford:

Matmulslow: CIoxf * 0.000057583kwh = 0.002706401 gCO2

Matmulfast: 0.007163599 gCO2

CFP for Singapore at 830pm:

Matmulslow: 0.028503585 gCO2

Matmulfast: 0.075446415 gCO2

CFP for singapore at 545am

Matmulslow: 0.025624435 gCO2

Matmulfast: 0.067825565 gCO2

As matrix operation is done billions of times everyday, even small per operation savings can have huge scale impacts. Running in a low CI region would be preferred.

Q14.

Energy per query: 0.34 Wh

Queries per day: 2.5 billion

Daily energy = 0.34 Wh × 2,500,000,000 = 850,000,000 Wh = 850 MWh per day

Given openai data centres are in texas and texas has CI of 400 gCO_2/kWh (https://www.dallasfed.org/-/media/documents/research/swe/2019/swe1903c.pdf)

Daily CFP = 850,000 kWh × 400 gCO_2/kWh = 340,000,000,000 gCO_2 = 340,000 tonnes CO_2 per day

Monthly CFP = 340,000 × 30 = 10,200,000 tonnes CO_2 per month

Yearly CFP = 340,000 × 365 = 124,100,000 tonnes CO_2 per year

https://medium.com/data-science/the-carbon-footprint-of-gpt-4-d6c676eb21ae

CFP of training GPT 4 is approx 13000tonnes of CO_2 (one off)

So chatgpt daily carbon footprint far exceeds the one off training carbon footprint.

An average car emits ~4 tonnes CO_2/year (https://www.nimblefins.co.uk/average-co2-emissions-car-uk). yearly ChatGPT inference ≈ 31 million cars

A lab of 20 desktop machines at ~100W each running 8hrs/day = 16 kWh/day → ChatGPT uses 53,000× more energy per day